

PRACTICE SET 3

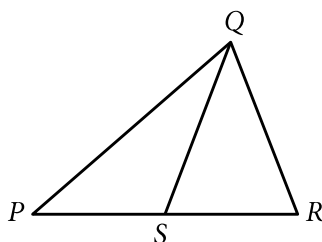
Quantitative Comparison Questions

For Questions 1 to 9, compare Quantity A and Quantity B, using additional information centered above the two quantities if such information is given. Select one of the following four answer choices and fill in the corresponding oval to the right of the question.

- Ⓐ Quantity A is greater.
- Ⓑ Quantity B is greater.
- Ⓒ The two quantities are equal.
- Ⓓ The relationship cannot be determined from the information given.

A symbol that appears more than once in a question has the same meaning throughout the question.

	<u>Quantity A</u>	<u>Quantity B</u>	<u>Correct Answer</u>
Example 1:	$(2)(6)$	$2 + 6$	Ⓐ Ⓑ Ⓒ Ⓓ



	<u>Quantity A</u>	<u>Quantity B</u>	<u>Correct Answer</u>
Example 2:	PS	SR	Ⓐ Ⓑ Ⓒ Ⓓ (since equal lengths cannot be assumed, even though PS and SR appear equal)

	$y < -6$		
	<u>Quantity A</u>	<u>Quantity B</u>	
1.	y	-5	Ⓐ Ⓑ Ⓒ Ⓓ

x is an integer and $23 < x < 27$.

- | | <u>Quantity A</u> | <u>Quantity B</u> | |
|----|---|-------------------|-----------------|
| 2. | The median of the five integers 23, 24, 26, 27, and x | 25 | (A) (B) (C) (D) |
-

r and t are consecutive integers and $p = r^2 + t$.

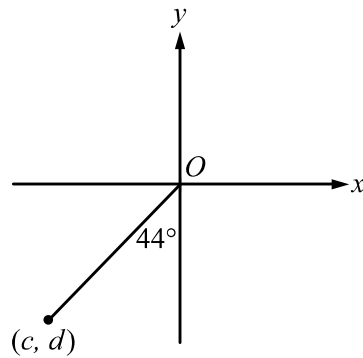
- | | <u>Quantity A</u> | <u>Quantity B</u> | |
|----|-------------------|-------------------|-----------------|
| 3. | $(-1)^p$ | -1 | (A) (B) (C) (D) |
-

The function f is defined by $f\left(\frac{x+3}{2}\right) = 3x^2 - x + 5$ for all numbers x .

- | | <u>Quantity A</u> | <u>Quantity B</u> | |
|----|-------------------|-------------------|-----------------|
| 4. | $f(4)$ | 75 | (A) (B) (C) (D) |
-

The sum of the annual salaries of the 21 teachers at School X is \$781,200. Twelve of the 21 teachers have an annual salary that is less than \$37,000.

- | | <u>Quantity A</u> | <u>Quantity B</u> | |
|----|--|---|-----------------|
| 5. | The average (arithmetic mean) of the annual salaries of the teachers at School X | The median of the annual salaries of the teachers at School X | (A) (B) (C) (D) |



- | | <u>Quantity A</u> | <u>Quantity B</u> | |
|----|-------------------|-------------------|---------|
| 6. | c | d | Ⓐ Ⓑ Ⓒ Ⓓ |
-

$N = 824^x$, where x is a positive integer.

- | | <u>Quantity A</u> | <u>Quantity B</u> | |
|----|---|-------------------|---------|
| 7. | The number of possible values of the units digit of N | 4 | Ⓐ Ⓑ Ⓒ Ⓓ |
-

$x > 0$ and $x \neq 1$.

- | | <u>Quantity A</u> | <u>Quantity B</u> | |
|----|-------------------|--------------------|---------|
| 8. | $(2x^{-4})(3x^2)$ | $\frac{24x}{4x^2}$ | Ⓐ Ⓑ Ⓒ Ⓓ |
-

- | | <u>Quantity A</u> | <u>Quantity B</u> | |
|----|--|--|---------|
| 9. | The length of a leg of an isosceles right triangle with area R | The length of a side of a square with area R | Ⓐ Ⓑ Ⓒ Ⓓ |

Multiple-choice Questions—Select One Answer Choice**For Questions 10 to 17, select a single answer choice.**

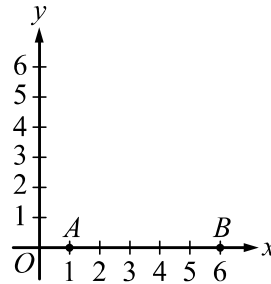
10. The relationship between temperature C , in degrees Celsius, and temperature F , in degrees Fahrenheit, is given by the formula $F = \frac{9}{5}C + 32$. If a recipe calls for an oven temperature of 210 degrees Celsius, what is the oven temperature in degrees Fahrenheit?
- Ⓐ 320
Ⓑ 350
Ⓒ 410
Ⓓ 420
Ⓔ 500
11. Of the students in a school, 20 percent are in the science club and 30 percent are in the band. If 25 percent of the students in the school are in the band but are not in the science club, what percent of the students who are in the science club are not in the band?
- Ⓐ 5%
Ⓑ 20%
Ⓒ 25%
Ⓓ 60%
Ⓔ 75%
12. Each year, the members of a book club select novels and nonfiction books to read. The club meets 3 times to discuss each novel and 5 times to discuss each nonfiction book they select. Last year, the club met 52 times and discussed 12 books. How many novels did the club discuss last year?
- Ⓐ 2
Ⓑ 4
Ⓒ 5
Ⓓ 7
Ⓔ 14

13. If $-1 < x < y < 0$, which of the following shows the expressions xy , x^2y , and xy^2 listed in order from least to greatest?
- Ⓐ xy, x^2y, xy^2
Ⓑ xy, xy^2, x^2y
Ⓒ xy^2, xy, x^2y
Ⓓ xy^2, x^2y, xy
Ⓔ x^2y, xy^2, xy
14. The 5 letters in the list G, H, I, J, K are to be rearranged so that G is the 3rd letter in the list and H is not next to G. How many such rearrangements are there?
- Ⓐ 60
Ⓑ 36
Ⓒ 24
Ⓓ 12
Ⓔ 6
15. If j and k are even integers and $j < k$, which of the following equals the number of even integers that are greater than j and less than k ?
- Ⓐ $\frac{k-j-2}{2}$
Ⓑ $\frac{k-j-1}{2}$
Ⓒ $\frac{k-j}{2}$
Ⓓ $k-j$
Ⓔ $k-j-1$

16. Which of the following is closest to $\sqrt{2.3 \times 10^9}$?
- (A) 50,000
 - (B) 150,000
 - (C) 500,000
 - (D) 1,500,000
 - (E) 5,000,000
17. The interior dimensions of a rectangular tank are as follows: length 110 centimeters, width 90 centimeters, and height 270 centimeters. The tank rests on level ground. Based on the assumption that the volume of water increases by 10 percent when it freezes, which of the following is closest to the maximum height, in centimeters, to which the tank can be filled with water so that when the water freezes, the ice would not rise above the top of the tank?
- (A) 230
 - (B) 235
 - (C) 240
 - (D) 245
 - (E) 250

Multiple-choice Questions—Select One or More Answer Choices

For Questions 18 to 19, select all the answer choices that apply.



18. Points A and B are shown in the xy -plane above. Point C (not shown) is above the x -axis so that the area of triangle ABC is 10. Which of the following could be the coordinates of C ?

Indicate all such coordinates.

- ☐ A (0, 4)
- ☐ B (1, 3)
- ☐ C (2, 5)
- ☐ D (3, 4)
- ☐ E (4, 5)

19. In a factory, machine A operates on a cycle of 20 hours of work followed by 4 hours of rest, and machine B operates on a cycle of 40 hours of work followed by 8 hours of rest. Last week, the two machines began their respective cycles at 12 noon on Monday and continued until 12 noon on the following Saturday. On which days during that time period was there a time when both machines were at rest?

Indicate all such days.

- ☐ A Monday
- ☐ B Tuesday
- ☐ C Wednesday
- ☐ D Thursday
- ☐ E Friday
- ☐ F Saturday

Numeric Entry Questions

For Questions 20 to 21, enter your answer in the answer box(es) below the question.

- Your answer may be an integer, a decimal, or a fraction, and it may be negative.
- If a question asks for a fraction, there will be two boxes—one for the numerator and one for the denominator. A decimal point cannot be used in a fraction.
- Equivalent forms of the correct answer, such as 2.5 and 2.50, are all correct. Fractions do not need to be reduced to lowest terms, though you may need to reduce your fraction to fit in the boxes.
- Enter the exact answer unless the question asks you to round your answer.

AVERAGE RATING OF PRODUCT X
GIVEN BY THREE GROUPS OF PEOPLE

Group	Number of People in Group	Average Rating
A	45	3.8
B	25	4.6
C	30	4.2

20. Each of the people in three groups gave a rating of Product X on a scale from 1 through 5. For each of the groups, the table above shows the number of people in the group and the average (arithmetic mean) of their ratings. What is the average of the ratings of the product given by the 100 people in the three groups combined?

Give your answer to the nearest 0.1.

21. The first term in a certain sequence is 1, the 2nd term in the sequence is 2, and, for all integers $n \geq 3$, the n th term in the sequence is the average (arithmetic mean) of the first $n - 1$ terms in the sequence. What is the value of the 6th term in the sequence?

Give your answer as a fraction.

Data Interpretation Set

Questions 22 to 25 are based on the following data. For these questions, select a single answer choice unless otherwise directed.

SIGHTINGS OF SELECTED BIRD SPECIES
IN PARK *H* IN 1999, BY SEASON

Species	Number of Sightings			
	Winter	Spring	Summer	Fall
Cardinal	30	18	11	20
Goldfinch	6	12	6	9
Junco	12	0	0	6
Nuthatch	8	2	0	4
Robin	6	12	28	18
Sparrow	20	19	23	22
Wren	0	18	30	12

22. In the winter, $\frac{2}{3}$ of the cardinal sightings, $\frac{1}{2}$ of the junco sightings, and $\frac{1}{4}$ of the sparrow sightings were in January. What fraction of the total number of sightings of these three bird species in the winter were in January?
- Ⓐ $\frac{1}{4}$
- Ⓑ $\frac{1}{3}$
- Ⓒ $\frac{1}{2}$
- Ⓓ $\frac{2}{3}$
- Ⓔ $\frac{3}{4}$
23. For which of the following bird species is the standard deviation of the numbers of sightings shown for the four seasons least?
- Ⓐ Cardinal
- Ⓑ Junco
- Ⓒ Robin
- Ⓓ Sparrow
- Ⓔ Wren

24. Which of the following is closest to the average (arithmetic mean) number of cardinal sightings for the 4 seasons?
- Ⓐ 12
 - Ⓑ 14
 - Ⓒ 16
 - Ⓓ 18
 - Ⓔ 20

For Question 25, use the directions for Numeric Entry questions.

25. By what percent did the number of wren sightings increase from spring to summer?

Give your answer to the nearest whole percent.

 %